Field	Value
Document No.	QP-005
Title	Production Planning and Control
Revision	B
Effective Date	2026-08-10
Prepared By	Owner
Approved By	Jacob Honick, Owner
Revision Approval Date	2026-11-05
Next Review	2027-08-10

QP-005 Production Planning and Control

Standards and clause references are guidance only and do not claim certification or standard compliance. Applicable customer, contractual and legal requirements govern accepted work.

1. Purpose

Define the process for planning, scheduling, and controlling production operations to ensure parts are manufactured in a controlled sequence with full traceability.

2. Scope

Applies to all machining operations performed on the two CNC machines and one manual mill in the shop, including operation sequencing, job traveler preparation, scheduling, and in-process traceability.

3. Responsibilities

Role	Responsibility
Shop Owner	Plan operations, create job travelers, schedule work across machines, maintain traceability records, and ensure production proceeds per the approved plan.

4. References

- QF-004 — Job Traveler / Route Sheet
- QF-005 — In-Process Inspection Record
- QP-002 — Contract Review
- QP-006 — In-Process Inspection
- QP-015 — Configuration Management
- QP-014 — Risk Management (operational risk assessment per AS9100D Clause 8.1.1)
- WI-004 — CNC Program Verification and Prove-Out
- AS9100 Rev D Clause 8.5.1

5. Definitions

- **Job Traveler:** The controlled document that accompanies the part through all production operations, recording what was done, by whom, and when.
- **Operation Sequence:** The planned order of manufacturing steps required to produce a conforming part from raw material to finished product.

6. Procedure

Job Planning

1. After contract review acceptance (QP-002), create a production plan for the job. 1a. Perform a job-level operational risk assessment per QP-014 Part B. Document risk factors on the Job Traveler (new part, new material, tight tolerances, compressed timeline, unfamiliar special processes). Apply additional controls for elevated-risk jobs.
2. Review the drawing and identify:
    - Raw material requirements (type, size, condition)
    - Machining operations required and their logical sequence
    - Special process requirements (heat treat, plating, NDT) and where they occur in the sequence
    - Inspection points (first piece, in-process, final)
    - Tooling and fixture requirements
    - CNC program requirements (existing or new program needed)

Job Traveler Creation

1. Prepare QF-004 Job Traveler for the job. Include:
    - Job number, customer name, part number, drawing number and revision
    - Material specification and lot/heat traceability reference
    - Operation sequence (numbered steps) with description of each operation

    - Machine assignment for each operation
    - Inspection hold points (operations requiring sign-off before proceeding)
    - Special process send-out points with supplier identification
    - Space for operator sign-off (initials, date) at each operation
2. Verify the traveler matches the drawing requirements and contract review notes.

Operation Sequencing

1. Sequence operations considering:
    - Geometric precedence (rough before finish, OD before ID where required)
    - Heat treatment timing (if required between machining operations)
    - Inspection points after critical dimensions are established
    - Outside processing send-out points (minimize handling and WIP time)
2. Identify which of the three machines each operation is assigned to based on capability and availability.

Scheduling

1. Schedule jobs considering:
    - Customer-required delivery date (from contract review)
    - Material availability (lead time for procurement)
    - Outside processing lead times
    - Current machine loading and job priorities
    - First Article Inspection time (if FAI is required)
2. Communicate delivery concerns to the customer as early as possible if scheduling conflicts arise.

Production Execution and Control

1. Release the job traveler and material to the shop floor only after:
    - Material has passed receiving inspection (QP-004)
    - CNC programs are verified per WI-004 (for new or revised programs)
    - Required tooling and fixtures are available
2. Perform each operation in the sequence documented on QF-004. Sign off each operation upon completion.
3. Do not skip or reorder operations without documented engineering authorization and updated traveler.

Traceability

1. Maintain lot/heat traceability from raw material through finished part:
    - Record material heat/lot number on the job traveler at the first operation.
    - If multiple lots are used in a batch, maintain lot segregation or document all lot numbers on each traveler.

2. For serialized parts (customer requirement), assign serial numbers at the first machining operation and record on the traveler.

Changes During Production

1. If a drawing change or process change is required during production, stop work and re-evaluate per QP-025 Control of Changes.
2. Update the job traveler to reflect any authorized changes. Note the change authorization reference on the traveler.

7. Records Generated

Form No.	Title	When Completed
QF-004	Job Traveler	Created at job planning, completed through production
QF-005	In-Process Inspection Record	At each designated inspection hold point

8. Revision History

Rev	Date	Description	Approved By
A	2026-08-07	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Standardize control metadata and annual review, remove fixed page counts and align applicable references and wording.